

Closing the gaps from Mock 1

A targeted study guide built from your first full 130-question mock. Your clinical reasoning is already strong; the points you are losing come from four trainable, mechanical skills. This guide teaches each one, drills it, and gives you a ten-study-day plan leading to Mock 2 on Oct 17.

78%

MOCK 1 · 101 / 130

65%

STUDY READINESS

72

DAYS TO EXAM · DEC 14

01 · Calculations

Time constant ↔ LFF, page-speed frequency and duration, sampling, 10-20, gain.

02 · Montage polarity

Bipolar subtraction, phase reversals, eye-movement dipoles, active references.

03 · ACNS 2021

Seizure, ESE, BIRDs and IIC thresholds, applied to numbers.

04 · Emergencies

Infant choking, AED with electrodes on, sedation airway, seizures.

Strong reasoning, leaky mechanics

You scored 101/130 (78%) on 130 higher-order Challenge questions, distributed exactly like the exam (19 / 60 / 25 / 26). The dashed line marks 80%, a comfortable margin above a typical pass threshold.

By exam domain

I · Pre-Study

74% 14/19

II · Performing

78% 47/60

III · Post-Study

76% 19/25

IV · Professional

81% 21/26

By skill

Troubleshooting

88% 21/24

Clinical

78% 49/63

Technical

72% 18/25

Montage

72% 13/18

Where the 29 missed points came from

AREA	YOUR ACCURACY	WHAT WENT WRONG	MODULE
Core calculations	46% · 28 answers	Time constant ↔ LFF (40%), frequency from non-standard page widths (0 of 3), samples per spike. Errors: decimal slips ×2, seconds read as hertz.	01
Polarity & montage	43% phase reversals	Referential → bipolar subtraction (0 of 3). Sign errors ×3 (blink, average reference, slow eye drift). Missed an active ear reference ×2.	02
Emergency response	1 of 3	Infant choking sequence, AED pad placement with EEG leads on, sedated child's airway.	04
ACNS 2021 thresholds	2 errors	3-Hz discharges for ≥10 s = seizure; triphasic GPDs at 1.5 Hz = IIC.	03

The pattern behind the numbers

Your accuracy rises with difficulty: **L3 57%**, L4 80%, L5 77%, **L6 87%**. The hardest reasoning questions are your best; the lost points sit in the most mechanical steps, such as a sign, a unit or a page width. That is good news: mechanics improve fastest with short, deliberate repetition.

How to use this guide

Work one module per study block: read the rules, cover the worked solutions and redo them yourself, then take that module's questions in the practice test (p. 10). In the app, use **Calculation Drill**, the **Repeated mistakes** drills and **Spaced review** daily.

Calculations without slips

Every calculation on the exam reduces to five relationships. Write the units at every step; most lost points were unit or decimal slips, not wrong formulas.

Amplitude (μV) = sensitivity ($\mu\text{V}/\text{mm}$) $\times$ deflection (mm)

A larger $\mu\text{V}/\text{mm}$ number means LOWER sensitivity: smaller waves.

speed (mm/s) = page width (mm) $\div$ page time (s)

Never assume 30 mm/s: compute it from the page every time.

duration (s) = width (mm) $\div$ speed $\cdot$ f (Hz) = 1 $\div$ period

Spike < 70 ms $\cdot$ sharp wave 70–200 ms.

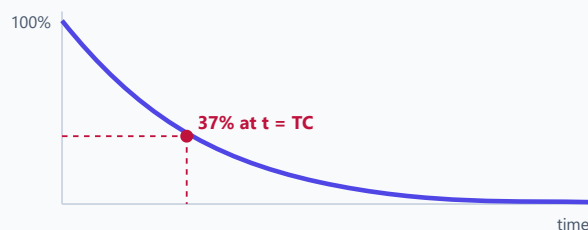
LFF = 1 $\div$ ($2\pi \cdot \text{TC}$) $\cdot$ TC = 1 $\div$ ($2\pi \cdot \text{LFF}$)

TC is the time to decay to 37% on a square-wave calibration.

samples = duration (s) $\times$ sampling rate $\cdot$ Nyquist = rate $\div$ 2

A filter passes $\sim 70\%$ (-3 dB) at its own cutoff.

Time-constant anchors: memorise these three



TIME CONSTANT	LFF
1 s	0.16 Hz
0.3 s	0.53 Hz
0.1 s	1.6 Hz

Shortcut: LFF $\approx 0.16 \div \text{TC}$. Halve the TC and the LFF doubles.

YOUR TRAPS ON MOCK 1

Decimal slip: TC 0.1 s gives 1.6 Hz, not 0.16 Hz.

Sanity-check against the anchors.

Seconds read as hertz: an LFF of 0.5 Hz is not a 0.5-s time constant; it is 0.32 s.

Assumed speed: a 20-s page across 300 mm runs at 15 mm/s, not 30.

Worked examples · cover the solution, then check

PAGE WIDTH

A 20-s review page is 300 mm wide. LPDs recur every 24 mm. What is their frequency?

- 1 Speed = $300 \text{ mm} \div 20 \text{ s} = 15 \text{ mm/s}$
- 2 Period = $24 \text{ mm} \div 15 \text{ mm/s} = 1.6 \text{ s}$
- 3 $f = 1 \div 1.6 \text{ s} = 0.63 \text{ Hz}$

At an assumed 30 mm/s you would report 1.25 Hz, double the truth.

DURATION

On a 10-s page 250 mm wide, a transient is 3 mm wide at its base. Spike or sharp wave?

- 1 Speed = $250 \div 10 = 25 \text{ mm/s}$
- 2 Duration = $3 \div 25 = 0.12 \text{ s} = 120 \text{ ms}$
- 3 70–200 ms → **sharp wave**

Assuming 30 mm/s gives 100 ms. Same class here, but not always.

TIME CONSTANT

A square-wave calibration decays to 37% after 0.2 s. What LFF is applied?

- 1 Decay to 37% → **TC = 0.2 s**
- 2 LFF = $1 \div (2\pi \times 0.2) = 1 \div 1.257$
- 3 = **0.8 Hz** (check: $0.16 \div 0.2 = 0.8 \checkmark$)

SAMPLING

A 30-ms spike is digitised at 200 samples/s. How many samples describe it?

- 1 Convert: 30 ms = **0.030 s**
- 2 $0.030 \text{ s} \times 200 \text{ /s} = 6 \text{ samples}$
- 3 Few samples can clip the true peak, so the amplitude may be underestimated

10-20 measuring: the percentages that matter

LINE	FROM THE START POINT
Midline nasion → inion	Fpz 10% · Fz 30% · Cz 50% · Pz 70% · Oz 90%
Coronal preauricular → preauricular	T3 10% · C3 30% · Cz 50% · C4 70% · T4 90%
Circumference through Fpz, T3, Oz	Fp1 5% · F7 15% · T3 25% · T5 35% · O1 45% from Fpz
Parasagittal	Fp1 → F3 → C3 → P3 → O1 at equal 25% steps of the Fp1–O1 distance (Fp1 10%, F3 30%, C3 50%, P3 70%, O1 90% of the nasion–inion line through C3)

10-20

Nasion–inion is 36 cm. How far is Fz from Pz?

- 1 Fz at 30%, Pz at 70% → 40% apart
- 2 $0.40 \times 36 \text{ cm} = 14.4 \text{ cm}$

Adjacent midline sites are 20% apart (7.2 cm here); Fz→Pz skips Cz.

Gain, sensitivity and filters

CALIBRATION ERROR

50 μV at 5 $\mu\text{V}/\text{mm}$ should give 10 mm. One channel deflects 12 mm.

$$\text{Error} = (12 - 10) \div 10$$

(expected) = 20% high

Divide by the expected value, not the observed one.

SENSITIVITY

A wave is 7 mm tall at 10 $\mu\text{V}/\text{mm}$. Sensitivity becomes 7 $\mu\text{V}/\text{mm}$.

$$\text{Amplitude} = 10 \times 7 = \mathbf{70 \mu V};$$

$$\text{new height} = 70 \div 7 = \mathbf{10 \text{ mm}}$$

The voltage never changes; only its drawing does.

70% RULE

HFF 35 Hz. A 35-Hz component is truly 60 μV .

$$\text{At the cutoff} \approx 70\% \text{ passes: } 0.7 \times 60 = \mathbf{\approx 42 \mu V}$$

Not 50% (that is -6 dB), not unchanged, not removed.

THE 20-SECOND HABIT

Before any page-based calculation, write the speed in the margin: **width $\div$ time = mm/s**. It takes 5 seconds and removes the most common Mock 1 error.

SANITY CHECKS

Alpha is 8–13 Hz; LPDs are usually 0.5–2.5 Hz; spikes are tens of ms. If an answer falls far outside the expected range, re-check the units before choosing it.

Nyquist and aliasing in one picture

$$\text{Nyquist} = \text{rate} \div 2$$

Represented faithfully (below rate $\div$ 2)

Aliases to a false lower frequency

Example: sampling 256 Hz $\rightarrow$ Nyquist 128 Hz.

To show activity up to 100 Hz you need more than 200 samples/s, plus a margin for the anti-alias filter.

Polarity you can trust

02

Three of your misses were the same mistake: a sign. Use one rule, applied the same way every time, and polarity questions become arithmetic.

Channel = Input 1 – Input 2

Negative result → pen goes **UP**. Positive result → pen goes **DOWN**.

Pens point TOWARD each other → negative maximum

Pens point AWAY from each other → positive maximum, at the shared electrode.

No reversal + biggest at the end of a chain

→ maximum at the end electrode (end-of-chain).
Confirm with another montage.

YOUR TRAPS ON MOCK 1

Adding instead of subtracting: $-60 - (-100) = +40$, not -160 .

Average reference sign: one electrode at $-380 \mu\text{V}$ (19 in the average) pulls the average to $-20 \mu\text{V}$, so every quiet electrode deflects **down** by $20 \mu\text{V}$.

Eye dipole: the cornea is **positive**. Wherever the eyes turn, the electrodes there go positive.

Worked chain · negative spike at T3

F7 $-30 \mu\text{V}$

F7-T3
 $-30 - (-110) = +80$
positive → DOWN 80



T3 $-110 \mu\text{V (max)}$

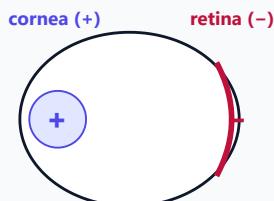
T3-T5
 $-110 - (-60) = -50$
negative → UP 50



T5 $-60 \mu\text{V}$

Pens point toward each other → negative max

Eye movements: the corneoretinal dipole



Gaze LEFT → cornea toward F7: F7 positive, F8 negative.

MOVEMENT	WHAT GOES POSITIVE	BIPOLAR PICTURE (NEGATIVE UP)
Blink (eyes roll up)	Fp1 and Fp2 together	Fp1-F3 and Fp2-F4 both down , in phase; no reversal (end of chain)
Look left	F7 (+), F8 (-)	F7-T3 down , F8-T4 up : out of phase
Look right	F8 (+), F7 (-)	F8-T4 down , F7-T3 up
Slow roving (drowsy)	Same as lateral gaze	Same signs, but slow (0.2–0.5 Hz) and no lateral rectus spikes

References: when the reference itself is active

Recognise it: every channel that shares one reference shows the *same* deflection, even electrodes far from the discharge; the true maximum can look small or inverted.

AVERAGE REFERENCE ARITHMETIC

19-electrode average. A pop drives O2 to +190 µV; all others are 0 µV. What do Cz-Avg and O2-Avg show?

1

Average = +190 ÷ 19 = **+10 µV**

2

Cz – Avg = 0 – 10 = -10 → **Cz up 10 µV**

3

O2 – Avg = 190 – 10 = +180 → **O2 down 180 µV**

Choosing a reference for bilateral temporal spikes involving both ears

STATE	BEST CHOICE	WHY THE OTHERS FAIL
Awake	Cz reference	Both ears carry spikes; Fpz is full of eye artifact. Cz is far from the temporal lobes and quiet while awake.
Asleep	Common average	Cz now carries vertex waves and K-complexes; the average dilutes any single event across all electrodes.
Any state	Bipolar chains	Reference-free: use them to confirm localisation with phase reversals.

REMEMBER THE TRADE-OFF

No reference is "true". Pick the one farthest from the activity you are studying, and confirm in a second montage.

Practice the subtraction: fill in, then check (answers at the end of the answer key)

REFERENTIAL VALUES (QUIET REFERENCE)	CHANNEL	YOUR CALCULATION	DIRECTION & SIZE
Fp2 -20 · F4 -90	Fp2-F4		
F4 -90 · C4 -40	F4-C4		
C3 +15 · P3 +55	C3-P3		
P3 +55 · O1 +10	P3-O1		

What do rows 1–2 and rows 3–4 each tell you about the location and sign of the maximum?

ACNS 2021: classify by the numbers

Each critical-care term is decided by two numbers: **frequency** (count discharges ÷ seconds) and **duration**. Work them out first, then read the table.

TERM	CRITERIA	HOW TO RECOGNISE IT FAST
Electrographic seizure	Epileptiform discharges averaging > 2.5 Hz for ≥ 10 s (> 25 in 10 s), or any pattern with definite evolution lasting ≥ 10 s	9 discharges in 3 s = 3 Hz; held for 15 s → seizure, even without evolution
Electrographic status (ESE)	Seizure lasting ≥ 10 continuous minutes , or seizures totalling ≥ 20% of any 60-minute period (≥ 12 min)	13 min of seizures within one hour → ESE, even if none lasted 10 min
BIRDS	Rhythmic activity > 4 Hz , at least 6 waves, lasting 0.5 to < 10 s . Definite : evolution or resembles the patient's seizures. Possible : sharply contoured without evolution.	7 s of 5 → 7 Hz speeding up = definite BIRDS (too short for a seizure)
Ictal-interictal continuum (IIC)	PDs or SW averaging > 1 and ≤ 2.5 Hz for ≥ 10 s ; or PDs/SW 0.5–1 Hz with a plus modifier or fluctuation; or LRDA > 1 Hz with a plus modifier or fluctuation	GPDs at 1.5 Hz with triphasic morphology in liver failure → IIC ; morphology and cause do not exclude it
LPDs / GPDs / LRDA / GRDA	Main terms: localisation (G, L, BI, UI, Mf) + pattern (PD, RDA, SW); plus modifiers +F (fast activity), +R (PDs only), +S (RDA only)	Asymmetric but synchronous PDs count as lateralized

Decision path

1. Frequency = discharges ÷ seconds · Duration = how long

2. > 2.5 Hz (or evolving) and ≥ 10 s? → **Seizure**

3. Seizure ≥ 10 min, or ≥ 12 of any 60 min? → **ESE**

4. > 4 Hz but < 10 s? → **BIRDS** (definite if evolving)

5. PDs/SW > 1 to 2.5 Hz for ≥ 10 s? → **IIC**

YOUR TRAPS ON MOCK 1

"Seizures must evolve." No: a steady > 2.5 Hz discharge for ≥ 10 s already qualifies.

"Triphasic waves are metabolic, not IIC." In ACNS 2021 they are GPDs with triphasic morphology, classified by frequency like any other PD.

TECHNOLOGIST'S ROLE

Measure frequency and duration, test and document responsiveness, and notify the physician promptly; the physician makes the clinical diagnosis.

Emergencies in the EEG lab

Emergency questions reward the first correct action. Learn each sequence as a fixed script (based on current AHA / Red Cross guidance; always follow your facility's protocol).

Choking infant (< 1 year), responsive

- 1 **5 back blows**, infant face-down along your forearm, head lower than chest
- 2 **5 chest thrusts**, face-up, heel of one hand on the lower half of the breastbone (AHA 2025 no longer prefers the two-finger technique)
- 3 Repeat until the object comes out or the infant becomes unresponsive → start **CPR** and call for help

No abdominal thrusts in infants; no blind finger sweeps.

Cardiac arrest with EEG electrodes on

- 1 **Keep CPR going** while the AED pads go on; do not pause for electrodes
- 2 A pad must not sit over an electrode or patch: **peel off the chest ECG electrode** where the pad belongs
- 3 **Scalp electrodes stay on**; they do not interfere with pad placement

Sedated child: snoring stops, SpO₂ falls

- 1 Effort without airflow means **obstruction**: open the airway with **head tilt–chin lift or jaw thrust**
- 2 **Call for help** at the same time: summon the sedation nurse/rapid response team
- 3 Oxygen helps only once the airway is open

Generalized convulsion during the EEG

- 1 **Time it** and keep recording; protect the head, clear the area
- 2 **Nothing in the mouth**; do not restrain the limbs
- 3 Afterwards, place the patient on their **side** (recovery position) and check the airway
- 4 Alert the nurse/physician at onset (per protocol); it is a medical emergency (status epilepticus) if it lasts **≥ 5 minutes**, recurs without recovery, or the patient is injured or not breathing normally

YOUR TRAPS ON MOCK 1

Choosing a reasonable but **later** step first (for example oxygen before opening the airway, or waiting before calling for help). On emergency questions ask: **"What keeps the patient alive in the next 30 seconds?"**

1 A 10-s page is displayed 250 mm wide. LPDs recur every 20 mm. What is their frequency?

- 2** A recording's time constant is 0.2 s. Which LFF does this correspond to?

- 3** The LFF is set to 1.6 Hz. What is the time constant?

- 4** A square-wave calibration decays to 37% of its height after 0.16 s. What LFF is applied?

- 5** A 40-ms spike is digitised at 256 samples per second. About how many samples fall within it?

- 6** A wave is 7 mm tall at 10 $\mu\text{V}/\text{mm}$. What is its amplitude, and how tall is it at 7 $\mu\text{V}/\text{mm}$?

- 7** The nasion-inion distance is 36 cm. How far apart are Fz and Pz along the midline?

- 8** The HFF is 35 Hz. A 35-Hz component has a true amplitude of 60 μV . About what amplitude is displayed?

9 Against a quiet reference: F7 $-30 \mu\text{V}$, T3 $-110 \mu\text{V}$, T5 $-60 \mu\text{V}$. What do F7-T3 and T3-T5 show (negative up)?

- 10** At the same instant F4-C4 deflects up and C4-P4 deflects down (negative up). What does this indicate?
- A** A negative maximum at C4
 - B** A positive maximum at F4
 - C** A positive maximum at C4
 - D** An electrode artifact at P4
- 11** Fp2-F8 shows a 60 μ V upward deflection, F8-T4 20 μ V up and T4-T6 5 μ V up; there is no phase reversal. Where is the maximum?
- A** Negative maximum at F8
 - B** Positive maximum at T6
 - C** No localisation is possible
 - D** Negative maximum at Fp2 (end of chain)
- 12** During a blink, what do Fp1-F3 and Fp2-F4 show (negative up)?
- A** Both deflect down, in phase
 - B** Both deflect up, in phase
 - C** Fp1-F3 up, Fp2-F4 down
 - D** No deflection; blinks cancel
- 13** The patient looks to the RIGHT. What do F7-T3 and F8-T4 show (negative up)?
- A** F7-T3 down, F8-T4 up
 - B** Both down
 - C** F7-T3 up, F8-T4 down
 - D** Both up
- 14** A 19-electrode average reference: O2 jumps to +190 μ V, all others stay at 0 μ V. What do Cz-Avg and O2-Avg show?
- A** Cz down 10; O2 down 190
 - B** Cz up 10; O2 down 180
 - C** Cz flat; O2 down 190
 - D** Cz up 10; O2 up 180
- 15** Left ear reference: during a left temporal spike, Fp1-A1, F3-A1, C3-A1 and O1-A1 all deflect down by a similar amount, while T3-A1 deflects slightly up. What explains this?
- A** A positive parasagittal field
 - B** Reversed leads in four channels
 - C** A1 lies in the spike field (active reference)
 - D** A negative maximum at C3
- 16** Independent spikes over both temporal regions involve both ears. Which references are best for a referential review awake, and then asleep with vertex waves?
- A** Linked ears; then linked ears
 - B** Cz; then Cz
 - C** Average; then Cz
 - D** Cz; then common average

MODULE 03 · ACNS 2021

- 17** Generalized spike-wave: 18 discharges in every 6 s, lasting 12 s, with no evolution. Classification?
- A** Ictal-interictal continuum
 - B** Electrographic seizure
 - C** Brief potentially ictal rhythmic discharges
 - D** Interictal GPDs
- 18** LPDs at 2 Hz for 30 minutes, no evolution, no clinical correlate. Classification?
- A** Electrographic status epilepticus
 - B** Electrographic seizure
 - C** Ictal-interictal continuum
 - D** BIRDs

- 19** A 7-s run of focal rhythmic sharp activity starts at 5 Hz and speeds to 7 Hz. Classification?
- A** Definite BIRDS **B** Electrographic seizure
C Ictal-interictal continuum **D** LRDA with plus modifier
- 20** Within one 60-minute period, seizures add up to 13 minutes; none lasts 10 continuous minutes. Classification?
- A** Not ESE: a 10-minute continuous seizure is required **B** Ictal-interictal continuum
C Electrographic status epilepticus **D** Recurrent BIRDS

MODULE 04 · EMERGENCIES

- 21** A 10-month-old suddenly cannot cry or cough but is still conscious. What do you do?
- A** Abdominal thrusts until the object comes out **B** A blind finger sweep, then back blows
C Start CPR immediately **D** 5 back blows, then 5 chest thrusts, repeated
- 22** A patient wearing 21 scalp electrodes and an ECG patch below the right clavicle arrests. CPR is under way when the AED arrives. Best approach?
- A** Peel off the right chest ECG patch, place the pads, leave scalp leads **B** Stop CPR and remove all scalp electrodes first
C Place the right pad over the ECG patch to save time **D** Disconnect the EEG headbox before any shock
- 23** A sedated 3-year-old's snoring stops; the chest still moves, there is no airflow, and SpO₂ falls to 89%. First action?
- A** Increase oxygen flow and keep watching **B** Open the airway (head tilt or jaw thrust) and call for help
C Call the child's name loudly and wait **D** Document the event and continue the study
- 24** A patient has a generalized convulsion during the EEG. Which plan is correct?
- A** Insert a bite block, then call for help **B** Restrain the limbs to prevent injury
C Time it, protect, nothing in the mouth; alert staff now, emergency if ≥ 5 minutes **D** Stop recording and wait 10 minutes before calling

Score yourself

SECTION	QUESTIONS	YOUR SCORE	NEXT STEP IF BELOW TARGET
Calculations	1–8	/ 8	Below 7: in-app Calculation Drill (20) twice this week
Montage polarity	9–16	/ 8	Below 7: redo Module 02 worked examples; Repeated-mistakes drill
ACNS 2021	17–20	/ 4	Below 4: re-read the decision path; Challenge Me · Clinical
Emergencies	21–24	/ 4	Below 4: recite the four scripts aloud daily for 3 days

Answer key with worked reasoning

Module 02 fill-in answers are at the end

- 1 **C** $250 \div 10 = 25 \text{ mm/s}$; $20 \text{ mm} \div 25 = 0.8 \text{ s period} \rightarrow \mathbf{1.25 \text{ Hz}}$. A assumes 30 mm/s; B reports the period as a frequency.
- 2 **A** $1 \div (2\pi \times 0.2) = 1 \div 1.257 \approx \mathbf{0.8 \text{ Hz}}$. B is $1/\text{TC}$ without 2π ; C reads seconds as hertz; D is a decimal slip.
- 3 **D** $\text{TC} = 1 \div (2\pi \times 1.6) = 1 \div 10.05 \approx \mathbf{0.1 \text{ s}}$ (anchor). A is $1/\text{LFF}$ without 2π ; B is the 0.16-Hz anchor; C is a decimal slip.
- 4 **B** 37% defines $\text{TC} = 0.16 \text{ s} \rightarrow 1 \div (2\pi \times 0.16) \approx \mathbf{1 \text{ Hz}}$. C forgets 2π ; A reads the TC as hertz.
- 5 **C** $0.040 \text{ s} \times 256 / \text{s} = 10.2 \rightarrow \mathbf{\text{about } 10}$. B and D are factor-of-ten slips from not converting ms to s.
- 6 **B** $10 \times 7 = \mathbf{70 \mu\text{V}}$; $70 \div 7 = \mathbf{10 \text{ mm}}$. A multiplies by the new setting; the voltage itself never changes.
- 7 **D** Fz 30% $\rightarrow$ Pz 70% = 40% of 36 cm = $\mathbf{14.4 \text{ cm}}$. A is one 20% step; C is nasion $\rightarrow$ Cz.
- 8 **A** $\approx 70\%$ passes at the cutoff (-3 dB): $0.7 \times 60 \approx \mathbf{42 \mu\text{V}}$. B uses 50%; C ignores the filter.
- 9 **B** $\text{F7-T3} = -30 - (-110) = +80 \rightarrow \mathbf{\text{down } 80}$; $\text{T3-T5} = -110 - (-60) = -50 \rightarrow \mathbf{\text{up } 50}$. Pens point toward each other: negative maximum at T3. C adds instead of subtracting.
- 10 **C** $\text{F4-C4 up} \rightarrow \text{C4 more positive than F4}$; $\text{C4-P4 down} \rightarrow \text{C4 more positive than P4}$. Pens point away from each other: **positive maximum at C4**.
- 11 **D** Largest deflection in the first channel, falling along the chain, no reversal: **end-of-chain negative maximum at Fp2**. Confirm with a referential or transverse montage. B also fits the signs, but the gradient is steepest at Fp2-F8 ($60 \mu\text{V}$) and flattest at T4-T6 ($5 \mu\text{V}$), so the field peaks at Fp2, not T6.
- 12 **A** The eyes roll up (Bell's phenomenon), so the positive corneas approach Fp1/Fp2; both are input 1 and positive $\rightarrow$ **both down, in phase**.
- 13 **C** Gaze right: the cornea (+) moves toward F8, the retina (−) toward F7. $\text{F8-T4 positive} \rightarrow \text{down}$; $\text{F7-T3 negative} \rightarrow \mathbf{\text{up}}$. Out of phase.
- 14 **B** $\text{Average} = 190 \div 19 = +10 \mu\text{V}$. $\text{Cz} - 10 = -10 \rightarrow \mathbf{\text{up } 10}$; $\text{O2} = 190 - 10 = +180 \rightarrow \mathbf{\text{down } 180}$.
- 15 **C** Identical deflections in every channel sharing A1 mean the **reference is active**. T3, the true maximum, is only slightly more negative than the contaminated A1.
- 16 **D** Awake: **Cz** is far from the temporal lobes and quiet. Asleep, Cz carries vertex waves and K-complexes, so the **common average** is least contaminated. Both ears are active.
- 17 **B** $18 \div 6 = 3 \text{ Hz} > 2.5 \text{ Hz}$, lasting $12 \text{ s} \geq 10 \text{ s} \rightarrow \mathbf{\text{electrographic seizure}}$, with or without evolution.
- 18 **C** $\text{PDs} > 1$ and $\leq 2.5 \text{ Hz}$ for $\geq 10 \text{ s}$ without evolution $\rightarrow \mathbf{\text{IIC}}$. Not a seizure (not $> 2.5 \text{ Hz}$, no evolution), so not ESE.
- 19 **A** $> 4 \text{ Hz}$, evolving, but only $7 \text{ s} (< 10 \text{ s}) \rightarrow \mathbf{\text{definite BIRDS}}$. LRDA is $\leq 4 \text{ Hz delta}$.
- 20 **C** $\geq 20\%$ of any 60-minute period ($\geq 12 \text{ min}$) qualifies: $13 \text{ min} \rightarrow \mathbf{\text{ESE}}$, even without a 10-minute seizure.

- 21 **D** Responsive infant: **5 back blows + 5 chest thrusts**, repeated. No abdominal thrusts in infants; no blind sweeps; CPR only if unresponsive.
-
- 22 **A** Continue CPR; a pad must not cover a patch, so **peel the ECG patch**; scalp electrodes do not interfere and stay.
-
- 23 **B** Effort without airflow = obstruction: **open the airway and call for help**. Oxygen cannot pass a closed airway.
-
- 24 **C** Time, protect, nothing in the mouth, no restraint; recovery position afterwards; **alert staff at onset**; ≥ 5 minutes = status/emergency, or sooner if injured or not breathing.
-

Module 02 fill-in answers

$Fp2-F4 = -20 - (-90) = +70 \rightarrow$ **down 70** · $F4-C4 = -90 - (-40) = -50 \rightarrow$ **up 50** → pens toward each other: **negative maximum at F4**.

$C3-P3 = 15 - 55 = -40 \rightarrow$ **up 40** · $P3-O1 = 55 - 10 = +45 \rightarrow$ **down 45** → pens away from each other: **positive maximum at P3**.

A plan that fits around work

About 40–60 minutes a day. Each day starts with the app's **Spaced review** (questions you missed come back on a schedule), then one focused block. The goal for Mock 2 is to move calculations above 75% and montage above 80%.

DAY 1

Module 01 rules

- Read pp. 3–5
- Redo the 4 worked examples cold
- App: Calculation Drill · 10

DAY 2

Page-speed & duration

- Practice test Q1–8
- App: Calculation Drill · 20
- Write the speed in the margin, every time

DAY 3

Module 02 rules

- Read pp. 6–7
- Fill-in subtraction table
- App: Repeated mistakes · polarity drill

DAY 4

Eye dipole & references

- Practice test Q9–16
- Sketch the cornea (+) for each gaze
- App: Challenge Me · Montage · 10

DAY 5

ACNS 2021

- Read p. 8, practice Q17–20
- Count discharges ÷ seconds on 5 examples
- App: Challenge Me · Clinical · 10

DAY 6

Emergencies

- Read p. 9; recite 4 scripts
- Practice Q21–24
- App: Today's Study

DAY 7

Mixed review

- Whole practice test again, timed (35 min)
- Redo every miss with its worked answer

DAY 8

Speed

- App: Calculation Drill · 20, target < 60 s each
- App: Spaced review (all due)

DAY 9

Light day

- Spaced review only
- Skim all four "traps" boxes

DAY 10 · OCT 17

Mock 2 (130)

- Full timed mock in the app
- Compare with Mock 1 by module
- Ask for the next targeted guide

RATE YOUR CONFIDENCE HONESTLY

In practice questions, tap **Sure / Unsure / Guess** truthfully. A confident miss is the most valuable signal: it marks a misconception and comes back first in tomorrow's review.

Where to find it in the app

ABRET Study → **Calculation Drill**, **Repeated mistakes** (Drill 5), **Spaced review**, **Challenge Me** with a skill focus, and the **Mock exam schedule** card (Mock 2 on Oct 17).